Qwen

A Dual-Phase Framework for Calibrating and Contractually Guaranteeing Safety in the Artemis-Cyboquatic Quantum Controller

Deconstructing the Baseline Risk Profile: The R ≈ 0.16 Challenge

The objective to reduce the Risk-of-Harm (RoH) associated with the LogicalQubitMap controller from its current value of approximately 0.16 necessitates a deep diagnostic of its constituent risks and their context within the broader ecosafety framework of Project Artemis . This baseline score, encapsulated in the KER triad of `K 0.90, E 0.90, R 0.16`, serves as both a quantitative target and a qualitative indicator of the controller's maturity and operational stability . The elevated RoH is not an abstract measure but a direct consequence of residual risks concentrated within the logical-qubit fabric, specifically arising from three coupled and critical failure modes: syndrome stream misinterpretation and re-encode timing logic, inaccurate node availability assumptions during failover, and under-modeled Radio Frequency (RF) and energy tail effects during error-correction bursts . Understanding these sources is the foundational step toward devising an effective mitigation strategy. The project's governance model explicitly treats configurations with higher-than-target RoH values as residing in "research-only" status, ensuring they do not enter production lanes until their safety properties have been rigorously validated and improved .

The primary driver of the high RoH is the inherent fragility of quantum information and the complexity of managing it at scale. Syndrome stream misinterpretation represents the first major axis of risk. In any quantum error correction (QEC) protocol, the process begins with the measurement of stabilizer generators to produce an error syndrome, which identifies the type and location of errors without collapsing the logical state [58] [63]. However, these measurements are imperfect; readout errors, detector inefficiencies, and crosstalk between qubits can lead to incorrect syndromes [16] [105]. A misinterpreted syndrome can trigger a faulty correction operation, inadvertently introducing a logical error where none existed or failing to correct an existing one. The second axis, node

availability and failover behavior, is equally critical. The LogicalQubitMap physically maps logical qubits onto physical segments distributed across a Quantum Virtual Machine (QVM) . When a physical segment fails, the controller must correctly identify it as unavailable and reconstruct the logical state using only the remaining healthy segments. If the controller makes an incorrect assumption about node health—for instance, by relying on stale telemetry or by misinterpreting transient noise as a permanent fault—it may attempt to use a compromised physical resource, leading to catastrophic logical state corruption [116]. This highlights the need for highly accurate and timely node health monitoring, a challenge complicated by the dynamic nature of quantum hardware environments.

The third, and perhaps most subtle, risk axis is the under-modeling of RF and energy tails. Superconducting qubits operate at cryogenic temperatures and are susceptible to disturbances from their electromagnetic environment [29]. High-energy operations, such as those performed during intensive error-correction cycles, generate significant RF plumes and Joule heating [30] [110]. These energy tails can propagate through the QVM's control lines and thermal links, potentially affecting neighboring qubits and violating operational constraints like the local RoH cap (`roh_cap_local`) or VECO (Vectorial Economic Capacity Overload) limit . An under-modeled energy kernel might predict a lower power draw or RF footprint than what actually occurs, causing the system to exceed its design envelopes during heavy computational workloads [8]. This can lead to temporary instability, increased decoherence rates, or even permanent hardware damage, all of which contribute to the overall Risk-of-Harm. The interplay between these three axes creates a complex risk landscape: a misinterpreted syndrome might trigger a re-encode event that places an undue load on a physically fragile node, whose own energy consumption was underestimated, pushing the entire region beyond its safe operating limits.

To contextualize the $R \approx 0.16$ score, it must be compared against the project's established safety benchmarks. The Artemis-Cyboquatic framework defines a target range for deployable configurations of $R \approx 0.12$–0.14 . Some production lanes may have even tighter caps, reflecting the highest levels of confidence in a system's safety properties . Therefore, a score of 0.16 situates the LogicalQubitMap controller firmly outside of this production-ready envelope, placing it in the same category as other unvalidated regimes that require further research and refinement before they can be safely integrated into the wider ecosystem . The corresponding Knowledge-factor ($K \approx 0.90$) suggests that while the controller is built upon proven principles and grammars from related domains, it has not yet undergone the full validation cycle required to achieve the higher K-values ($\approx$0.93–0.95) characteristic of mature, trusted components . Similarly, the Eco-impact

score (E ≈ 0.90) indicates a balance between functionality and efficiency, but the elevated RoH implies that there is room for improvement in how the controller manages its resources to minimize negative environmental and operational impacts . The strategic imperative is thus clear: to reduce R from 0.16 toward the 0.13 band targeted for deployable configurations, this must be achieved without degrading the existing E and K scores, which are considered acceptable starting points for this refinement effort .

This problem is situated within the overarching VSC-ARTEMIS spine, which mandates that all safety improvements must be both measurable and contractually immutable . The chosen metric for quantifying risk is the Lyapunov residual, V_t, a weighted sum of squared normalized risk coordinates (r_j) that provides a single scalar value representing the system's overall state of stability [32] [144]. For the LogicalQubitMap, the relevant residual would be defined as $V_{\text{logic},t} = \sum_j w_j r_j$, where the coordinates r_j could include metrics such as `nodeavailability`, `syndromeweight`, `agesincelastfullreencode`, `rfexposure`, and `energymargin` . The core principle governing the controller is that any allowed operation, such as a re-encode or a failover, must not increase this residual, i.e., $V_{\text{logic},t+1} \leq V_{\text{logic},t}$. By calibrating the individual risk coordinates r_j more accurately and tightening their corridor bands based on real-world pilot data, the controller's sensitivity to incipient failures increases. This allows it to take corrective action, like initiating a re-encode, earlier in the failure timeline, thereby preventing a small issue from escalating into a logical state loss. This directly addresses the user's goal of reducing both state-loss risk and the "thrash" risk associated with repeated, failed correction attempts . The table below summarizes the baseline risk profile and its implications.

KER Metric	Baseline Value	Target / Context	Implication
Knowledge-factor (K)	0.90	Production Ready: K ⩾ 0.90	The controller is based on sound principles but lacks full validation under real-world conditions.
Eco-impact (E)	0.90	Production Ready: E ⩾ 0.90	The controller is efficient but has room for improvement in resource management to lower negative impacts.
Risk-of-Harm (R)	0.16	Production Ready: R ⩽ 0.13	The controller is currently in "research-only" status due to residual risks dominated by error correction and scheduling.

Ultimately, the challenge is not merely to lower a number but to fundamentally improve the reliability and predictability of the quantum logical fabric. The three prioritized risk axes—syndrome interpretation, node availability, and energy modeling—are the levers through which this improvement must be achieved. Success will depend on moving from theoretical models to empirically calibrated ones, leveraging pilot telemetry to tighten the corridors within which the system operates. This initial diagnostic establishes the

foundation for the dual-phase strategy, which combines immediate tactical refinements with long-term structural guarantees, ensuring that the resulting reduction in RoH is both robust and enduring.

Phase 1 Implementation: Near-Term Tactical Refinements and Calibration

The first phase of the strategy to reduce the LogicalQubitMap controller's Risk-of-Harm (RoH) from 0.16 to the target of 0.13 is centered on near-term tactical refinements and empirical calibration . This approach leverages existing knowledge and applies it more precisely through the analysis of pilot telemetry data. The core mechanism for this phase is the enhancement of the Lyapunov-style residual controller, which governs the stability of the logical qubit fabric . By tightening the operational boundaries—the corridor bands—around the key risk coordinates that constitute the residual $V_{\text{logic},t}$, the controller becomes more sensitive to incipient failures. This heightened sensitivity enables earlier intervention, allowing the system to re-encode or perform failovers before minor issues escalate into catastrophic logical state loss or thrashing. The three prioritized axes of focus for this calibration are syndrome stream interpretation, node availability modeling, and RF/energy tail effects, as specified in the research goal .

The primary tool for this refinement is the calibration of each risk coordinate (r_j) within the logical residual. The residual is defined as $V_{\text{logic},t} = \sum_j w_j r_j$, where each r_j represents a normalized risk score for a specific domain, such as `nodeavailability`, `syndromeweight`, `agesincelastfullreencode`, `rfexposure`, and `energymargin` . Calibration involves using real pilot telemetry to determine more accurate distributions and thresholds for these coordinates. For example, regarding syndrome stream interpretation, calibration would involve analyzing vast datasets of parity check outcomes from pilot runs. This analysis would help distinguish between statistical noise, systematic biases in the measurement apparatus, and true underlying error patterns. The goal is to refine the `syndromeweight` coordinate so that it more accurately reflects the probability and severity of errors detected. This could involve developing more sophisticated decoding algorithms that account for known correlations in qubit errors, which are a well-documented phenomenon in superconducting devices [24] [129]. By improving the fidelity of the syndrome signal, the controller can make more confident and accurate decisions about when and how to correct errors, directly reducing the risk of applying a faulty correction.

Similarly, for node availability and failover behavior, calibration requires correlating telemetry from the LogicalQubitMap with low-level diagnostics from the Quantum Virtual Hardware Abstraction Layer (VHAL). This involves tracking metrics like two-qubit gate fidelities, coherence times, and readout errors for each physical qubit over time [116]. By observing how these metrics degrade, the `nodeavailability` coordinate can be updated to reflect a more realistic assessment of a segment's health. This prevents the controller from making decisions based on outdated or incomplete information. For instance, if a physical segment experiences a temporary dip in performance due to a cosmic ray event, a poorly calibrated system might mark it as permanently failed, whereas a calibrated system would monitor its recovery. Tightening the corridor bands around `nodeavailability` ensures that the controller only relies on nodes that have demonstrated sustained health, thereby preventing the reconstruction of logical states on compromised physical hardware. This process mirrors techniques used in federated calibration, where residual statistics are aggregated to estimate global drift patterns and mitigate localized errors [33].

The third axis, RF/energy tail modeling, requires a similar empirical approach. The contribution of a given quantum job to the `rfexposure` and `energymargin` coordinates depends on the accuracy of the underlying kernels for joules-per-gate and RF power consumption . These kernels are often initially estimated based on manufacturer specifications or simplified models, which can be inaccurate under heavy operational loads [9]. Pilot telemetry from high-duty-cycle experiments can provide the necessary data to recalibrate these kernels. By measuring the actual power draw and RF emissions during various gate sequences and error-correction cycles, a more precise relationship between circuit depth, gate count, and resource consumption can be established. This refined data tightens the corridor bands for the energy-related coordinates, allowing the scheduler to more accurately predict whether a quantum job will push its host region beyond its RoH or VECO caps. Implementing adiabatic scheduling, where gate batches are checked against a nominal change in the residual (`dvtnominal`) before execution, further hardens this protection, mirroring the robust control strategies seen in other complex systems [69].

Beyond refining the risk coordinates themselves, another critical tactical lever is the downscaling of static risk caps, particularly `rohcap_static` and `kfestimatestatic` for production LogicalQubitMaps . The `rohcap_static` field represents a hard upper bound on the residual risk that a logical map can tolerate. By lowering this cap, the controller is forced to initiate a re-encode at an earlier stage of potential degradation. This trades a marginal increase in compute and energy overhead (a small hit to E) for a significant gain in safety (a larger reduction in R) . This is a classic safety-critical trade-

off, analogous to the project's policy for biodegradable substrates. If a substrate's decomposition kinetics are slower than promised, it is treated as having a higher toxicity risk (`rtox↑`) and is moved to a more controlled, research-only environment until its behavior is better understood and constrained . Applying the same logic here, forcing the quantum controller to re-encode sooner is a proactive measure that reduces the window of vulnerability. The `kfestimatestatic` field, representing a static estimate of the Knowledge-factor, can also be tightened. A lower value acts as a conservative proxy for less certainty about the controller's long-term stability, again encouraging more frequent interventions to maintain safety margins. Together, these adjustments create a more aggressive but safer operational posture for production systems.

The implementation of these Phase 1 changes must be governed by the same rigorous standards as the rest of the Artemis-Cyboquatic stack. Every change to corridor bands, Lyapunov weights, or static caps must be encoded in a versioned ALN document, providing a clear audit trail of the rationale and evidence behind the change . Furthermore, the Continuous Integration (CI) pipeline should be augmented to automatically verify that any proposed configuration change still adheres to the fundamental invariant of a non-increasing Lyapunov residual ($V_{\text{logic},t+1} \leq V_{\text{logic},t}$) [3] . This can be done by simulating the effects of new jobs or changing corridor bands on the residual and ensuring that no violation is possible. The output of this verification process should be emitted as a `ResponseShard`, a tamper-evident record of the safety check, reinforcing the principle of "no corridor, no compile" . This combination of empirical calibration, parameter downscaling, and automated verification forms a powerful, multi-layered approach to immediately and measurably reduce the RoH of the LogicalQubitMap controller, paving the way for the more profound structural changes planned for Phase 2.

Phase 2 Governance: Long-Term Structural Guarantees and Formal Orthogonality

While Phase 1 focuses on tactical refinements and empirical tuning, Phase 2 of the strategy is dedicated to establishing long-term, structural guarantees that will lock in the safety improvements and prevent future regressions. This phase moves beyond parameter adjustment to encompass formal methods, enhanced system architecture, and immutable contractual enforcement. The primary objectives of Phase 2 are twofold: first, to introduce new, explicit quantum risk coordinates into the Lyapunov residual $V_{\text{logic},t}$ to capture previously unmodeled or insufficiently modeled aspects of quantum system

dynamics; and second, to develop and embed formal proofs of orthogonality that cryptographically guarantee the separation between the quantum control plane and the econet reward plane . These measures ensure that the reduced Risk-of-Harm is not just a result of good calibration but is a fundamental, verifiable property of the system's design.

A key initiative for Phase 2 is the introduction of new quantum risk coordinates if refined models from Phase 1 still leave residual uncertainty. The current set of coordinates—such as `nodeavailability`, `syndromeweight`, and `rfexposure`—while comprehensive, may not fully capture all sources of risk in a complex, large-scale quantum processor . One promising addition is an explicit "re-encode churn" coordinate. Re-encoding, while necessary for fault tolerance, is a disruptive and resource-intensive process. Excessive or unnecessary re-encoding events can lead to thrashing, where the system spends more time correcting itself than performing useful computation, ultimately degrading performance and increasing wear on the hardware. By defining a coordinate that tracks the frequency and cost of re-encoding events, the controller gains a direct measure of this operational friction. This allows the Lyapunov function to penalize configurations or policies that lead to excessive churn, encouraging a more balanced approach to error correction that minimizes both logical state loss and operational disruption.

Another valuable coordinate is a "syndrome model confidence" axis. As discussed in Phase 1, syndrome misinterpretation is a primary source of risk. While calibration improves the raw accuracy of syndrome detection, it cannot eliminate all uncertainty. A confidence coordinate would quantify the degree of trust the controller has in its current syndrome interpretation model. This confidence could be dynamically adjusted based on several factors, such as the consistency of syndromes over time, the presence of correlated error signatures, or the results of periodic sanity-checking circuits. A low confidence reading would act as a flag, potentially triggering a more conservative error-correction strategy (e.g., repeating measurements, seeking external validation) or elevating the system's perceived risk level. Incorporating such a coordinate makes the controller's internal reasoning more transparent and auditable, transforming it from a "black box" optimizer into a more interpretable safety instrument. The development of such coordinates aligns with efforts in AI-driven post-quantum cryptography, where concepts like confidence estimation are crucial for reliable system behavior [37]. The introduction of these new coordinates would be formally defined in ALN documents, making them first-class citizens in the system's safety grammar.

The second, and arguably more critical, component of Phase 2 is the creation of formal proofs and ALN-level contracts that enforce the architectural non-interference between the quantum plane and the econet reward plane . This requirement stems from the need to ensure that the RoH of the LogicalQubitMap controller, even when minimized, cannot

be used to negatively impact the accumulation of rewards in self-parasitic regeneration regions or elsewhere in the eco-wealth ecosystem . The concern is that if the quantum risk calculation were improperly exposed to the reward engine, a high-RoH event in the quantum controller could be translated into a negative multiplier on an Augmented Citizen's `ecoimpactscore` or `vampiric-health` attribute. This would violate the project's core principle of separating ecosafety from financial or reward-based incentives, creating perverse incentives and undermining trust .

To address this, Phase 2 will formalize the separation boundary. This involves writing precise ALN contracts that state unequivocally: "LogicalQubitMap RoH may gate admission and re-encode decisions, but MUST NOT be used as a negative multiplier on `ecoimpactscore`, `neurorightsstatus`, or any `vampiric-regeneration` attributes." These contracts are not mere suggestions; they are enforced as part of the ALN verification process, which governs all state transitions in the system [1]. Any transaction or policy update that attempts to violate this orthogonality would be rejected by the ALN validator before it can ever be compiled or deployed, providing compile-time assurance of the separation . This elevates the requirement from a runtime best practice to a constructively enforced invariant.

Furthermore, this orthogonality will be supported by formal proofs. Using mathematical proof assistants or symbolic execution tools, it will be possible to prove that the reward calculation functions, as encoded in the ALN contracts, are mathematically independent of the fields and residuals originating from the quantum plane. This proof becomes part of the system's constitution, a machine-verifiable artifact that demonstrates the correctness of the separation. This approach is inspired by the concept of modular minimalism in grammar, where distinct modules have clearly defined interfaces and interactions, preventing unintended cross-talk [2]. By treating the quantum resilience lane and the econet reward lane as separate, orthogonal modules, the system's overall complexity is managed, and its security properties become provable rather than merely assumed. This commitment to formal guarantees is central to the VSC-ARTEMIS spine's philosophy of making safety improvements contractually immutable . The successful completion of Phase 2 will mean that the Artemis-Cyboquatic stack can confidently integrate quantum computing capabilities, knowing that its newfound power is contained within a rigorously defined and provably secure boundary, protecting the integrity of the entire ecosafety ecosystem.

Architectural Enforcement: Shard-Lane Separation and ALN Contracts

The mandate for strict architectural non-interference between the quantum-plane RoH and the econet reward systems requires a multi-layered defense strategy that enforces separation at the data, logic, and code levels. This is not merely a matter of software design but a foundational requirement for maintaining the integrity of the eco-wealth economy, particularly in sensitive regions tied to self-parasitic regeneration processes . The proposed solution hinges on two primary mechanisms: the use of dedicated shard lanes to create logical data boundaries and the deployment of ALN contracts to enforce cryptographic rules at the point of transaction validation. Together, these mechanisms form a robust, defense-in-depth architecture that prevents the contamination of reward calculations by quantum risk metrics.

The first layer of enforcement is logical data separation through shard-lane partitioning. In the Artemis-Cyboquatic architecture, all state and activity are recorded in DID-signed, hex-stamped qpudatashards . Each shard contains fields for metrics like risk coordinates (`rx`), Lyapunov residual (`Vt`), and the KER triad . To enforce the required isolation, all shards originating from the quantum plane—including those from the `QuantumVMRegion`, `QuantumCircuitProfile`, and the `LogicalQubitMap` controller itself—will be assigned to a dedicated shard lane. A suitable name for this lane would be `QVM-LOGIC-RESILIENCE` . This lane acts as a clear, unambiguous container for all quantum-related safety and performance data. The existence of this distinct lane creates a strong convention and a practical barrier: any query, analysis, or computation that needs to operate on quantum data must explicitly target this lane. It becomes structurally impossible to accidentally join data from the `QVM-LOGIC-RESILIENCE` lane with data from the `NORMAL` or `SELF-PARASITIC-REGEN` lanes without an explicit, deliberate, and auditable cross-shard operation.

This shard-lane separation is complemented by a second, more powerful layer of enforcement: ALN contracts. While a shard lane provides a logical boundary, it can be bypassed by a sufficiently motivated or flawed program. ALN contracts provide a cryptographic and computationally enforced boundary. As outlined in the research goal, ALN specifications will be created that contain predicates forbidding any econet reward kernel from importing quantum residuals or RoH values as negative multipliers . For example, an ALN contract governing the calculation of an `ecoimpactscore` would include a rule stating that its formula cannot reference any fields present in a `LogicalQubitMap` shard, such as `roh_cap_static` or the calculated `r_logic` coordinates . When a transaction attempts to update a reward score, the ALN virtual

machine will execute these predicates as part of its verification process. If the transaction violates the orthogonality constraint, the entire transaction is rejected, regardless of the runtime environment's state. This transforms the isolation requirement from a soft constraint ("don't mix the data") into a hard, unbreakable rule ("this operation is forbidden by the system's constitution"). This compile-time enforcement, summarized by the principle "no corridor, no build," ensures that the fundamental separation of concerns is preserved by construction, not just by convention or careful coding .

The third and final layer of enforcement is prevention at the development stage through schema-level checks and CI/CD gates. Before any code change is merged, the CI pipeline will run automated tests that statically analyze the schemas of the involved shards and the logic of the ALN contracts . Specifically, the pipeline will check for cross-plane dependencies. For instance, it will verify that the schema for an `EconetRewardShard` has no declared fields that are aliases or derivatives of fields from a `QuantumLogicShard`. This static analysis acts as a first line of defense, catching potential violations of orthogonality before they are ever committed to the codebase. This practice aligns with modern software engineering principles that prioritize finding bugs early in the development lifecycle [33]. By combining these three layers—a clear data boundary via shard lanes, an unbreakable rule enforcement via ALN contracts, and a proactive prevention via CI/CD schema checks—the Artemis-Cyboquatic stack can establish a provably secure boundary between the high-risk quantum control plane and the stable econet reward plane. This layered approach ensures that even if one layer of defense is compromised or a developer makes an error, the other layers remain to protect the integrity of the entire system, fulfilling the user's demand for strict architectural non-interference.

Technical Blueprint: Repository Architecture and Key Object Definitions

To implement the phased strategy for reducing the Risk-of-Harm of the LogicalQubitMap controller and enforcing its architectural isolation, a clear and modular technical blueprint is required. This blueprint is grounded in the existing architecture of the Artemis-Cyboquatic stack, emphasizing the reuse of proven Rust crates and ALN specifications to maintain consistency and leverage the project's existing ecosafety grammar . The repository architecture should be organized around three core layers: the shared ecosafety math kernel, the Cyboquatic mechanics, and the specialized Artemis quantum-circuit plane, with all components emitting standardized qpudatashards and

being governed by ALN contracts . This structure ensures that the quantum plane is treated as just another corridor-governed domain under the same universal KER and Lyapunov framework used throughout the system .

The proposed top-level repository layout provides a clean separation of concerns and guides future development . The `crates/ker_core/` directory will house the central ecosafety mathematics. This crate will contain the definitions for `CorridorBands`, `MetricFields`, and the core logic for calculating the Lyapunov residual $V_t = \sum_j w_j r_{j,t}^2$. By centralizing this logic, all planes—from hydraulics to quantum circuits—can share a single, consistent kernel for risk normalization and residual computation, ensuring mathematical homogeneity across the entire stack . The `crates/qpu_shards/` directory will define the Rust structs that mirror the quantum-specific qpudatashards, including `QuantumVMRegion`, `QuantumCircuitProfile`, and the `LogicalQubitMap` object itself . These structs serve as the strongly-typed schema for the data exchanged between the quantum controller and the rest of the Artemis system.

The `crates/logical_qubits/` crate is the heart of the implementation. It will contain the Rust error-correction controller that implements the Lyapunov-style logic over syndrome streams and node health . This controller will consume telemetry from the quantum hardware, calculate the `r_j` coordinates, update the residual $V_{\text{logic},t}$, and decide whether to permit a job, schedule a re-encode, or initiate a failover, all while enforcing the invariant $V_{\text{logic},t+1} \leq V_{\text{logic},t}$. This crate will depend on `ker_core` for its mathematical functions and on `qpu_shards` for its data structures. Finally, the `crates/aln_specs/` directory will hold the versioned `.aln` files that serve as the system's constitution. These documents will define the particles, their required fields, corridor bands, validity conditions, and governance flags for every component, from the quantum VM regions to the FOG routing policies . This tight coupling between machine-readable ALN specs and typed Rust structs is a cornerstone of the project's design, ensuring that the human-readable documentation always corresponds to the executable logic .

Concrete Rust objects and traits, defined in alignment with the provided context, give this blueprint a tangible form. The `MetricFields` struct, residing in `ker_core`, provides the universal data format for reporting risk and residual information .

```
// crates/ker_core/src/lib.rs
#[derive(Clone, Debug)]
pub struct MetricFields {
    pub rx: Vec<f64>,   // Normalized risk coordinates
    pub vt: f64,       // Residual before transition
```

```
    pub vt_next: f64,  // Residual after transition
    pub k: f64,        // Knowledge-factor
    pub e: f64,        // Eco-impact
    pub r: f64,        // Risk-of-Harm
}
```

For the quantum plane, the `LogicalQubitMap` struct is a first-class ALN particle that binds logical qubits to physical hardware and carries its own static risk estimates .

```
// crates/logical_qubits/src/lib.rs
#[derive(Clone, Debug)]
pub struct PhysicalSegment {
    pub node_id: String,
    pub vm_id: String,
    pub qubit_index: u32,
    pub role: SegmentRole,
}

#[derive(Clone, Debug)]
pub struct LogicalQubitMap {
    pub logical_id: String,
    pub physical_segments: Vec<PhysicalSegment>,
    pub roh_cap_static: f64,     // Static RoH capacity cap
    pub kf_estimate_static: f64, // Static Knowledge-factor estimate
    pub veco_cap_static: f64,    // Static VECO capacity cap
    pub code_distance: u32,
    pub stabilizer_layout: String,
    pub syndrome_rate_hz: f64,
}
```

These objects, along with their counterparts for `QuantumVMRegion` and `QuantumCircuitProfile`, provide a concrete, type-safe surface for all future quantum-circuit research and development . Every new corridor, proof, or dataset becomes an ALN-versioned struct, preventing the proliferation of free-floating parameters and ensuring that all research contributes directly to tightening the system's safety envelope . The following table outlines the key ALN particles and their relevance to the RoH reduction goal.

ALN Particle	Purpose	Fields Relevant to RoH Reduction
`QuantumVMRegion`	Defines the properties and constraints of a QPU region.	`max_qubits`, `max_circuit_depth`, `joules_per_gate_base`, `roh_cap_local`, `veco_cap_local`, `rf_power_cap_local`
`QuantumCircuitProfile`	Specifies the resource budget and policies for a quantum job.	`roh_budget_job`, `veco_budget_job`, `finance_forbidden`, `trade_forbidden`, `weapons_forbidden`
`LogicalQubitMap`	Maps logical qubits to physical hardware segments.	`roh_cap_static`, `kf_estimate_static`, `veco_cap_static`, `syndrome_rate_hz`
`QuantumLogicShard`	A qpudatashard logging the state of the logical qubit fabric.	`r_nodeavailability`, `r_syndromeweight`, `r_agesincelastreencode`, `r_rfexposure`, `r_energymargin`, `vt_logic`

By building upon this solid technical foundation of modular Rust crates and a constitution-like ALN specification, the Artemis-Cyboquatic project can systematically and safely evolve its quantum capabilities, ensuring that every advance in computational power is matched by an equivalent or greater advance in ecosafety.

Synthesis and Strategic Path Forward

This research report has detailed a comprehensive, dual-phase strategy for reducing the Risk-of-Harm (RoH) of the LogicalQubitMap controller from its baseline of 0.16 to the target of 0.13, while simultaneously enforcing strict architectural isolation from the econet reward systems. The proposed path forward is a direct application of the core tenets of the VSC-ARTEMIS spine: leveraging calibrated corridors and Lyapunov residuals for immediate risk reduction, followed by locking these improvements into place with immutable, shard-level contracts and formal proofs. The entire endeavor is structured to ensure that the integration of quantum computing capabilities into the Artemis-Cyboquatic stack enhances its ecosafety posture rather than compromising it.

The first phase of this strategy consists of near-term tactical refinements. This involves calibrating the logical Lyapunov residual, $V_{\text{logic},t}$, by tightening the corridor bands for its constituent risk coordinates—`nodeavailability`, `syndromeweight`, `agesincelastfullreencode`, `rfexposure`, and `energymargin`—using real pilot telemetry . This empirical approach increases the controller's sensitivity to incipient failures, enabling earlier and more effective interventions such as re-encoding or failovers. Complementing this, the phase includes the downscaling of static risk caps like `roh_cap_static` to proactively force more frequent, albeit less disruptive, re-encoding events, thereby mitigating both state-loss and thrash risks . These actions directly address

the root causes of the elevated RoH and move the controller's KER score closer to the deployable threshold.

The second phase transitions from tactical tuning to long-term structural guarantees. This involves the introduction of new, explicit quantum risk coordinates, such as "re-encode churn" and "syndrome model confidence," to make the controller's reasoning more transparent and its risk management more robust . More critically, this phase focuses on creating formal proofs and ALN-level contracts that enforce the architectural non-interference between the quantum control plane and the econet reward plane . By assigning quantum-related data to a dedicated `QVM-LOGIC-RESILIENCE` shard lane and embedding ALN predicates that cryptographically forbid the use of quantum residuals in reward calculations, the project establishes a provably secure boundary . This ensures that the RoH of the quantum controller can never be used to unfairly penalize users in self-parasitic regeneration regions or any other part of the eco-wealth ecosystem, upholding the project's foundational principle of separating ecosafety from economic incentives.

In summary, the strategic path forward is clear and methodical. It begins with data-driven calibration to tune the system's sensitivity, then culminates in the construction of formal, unchangeable rules that govern its behavior. The technical blueprint, based on a modular Rust architecture and governed by versioned ALN specifications, provides the necessary tools to implement this plan with precision and verifiability. Successfully executing this strategy will not only lower the RoH of the LogicalQubitMap controller but will also serve as a powerful demonstration of the extensibility and robustness of the Artemis-Cyboquatic ecosafety spine, proving its ability to securely integrate next-generation technologies while advancing the mission of creating a provably non-harmful and globally trustworthy infrastructure.

Reference

1. [PDF] Contracts for Systems Design: Methodology and Application cases https://inria.hal.science/hal-01178469/document
2. [PDF] Realizational Morphology in a Modular Minimalist Grammar https://aclanthology.org/2025.mol-1.1.pdf

3. [PDF] Automated Static Vulnerability Detection via a Holistic Neuro ... - arXiv https://arxiv.org/pdf/2504.16057
4. Arxiv今日论文| 2025-12-24 - 闲记算法 http://lonepatient.top/2025/12/24/arxiv_papers_2025-12-24
5. 计算金融/密码安全/新兴技术2025_5_20 - arXiv每日学术速递 https://arxivdaily.com/thread/67592
6. Quantum Machine Learning for 6G Space-Air-Ground Integrated ... https://www.researchgate.net/publication/395590858_Quantum_Machine_Learning_for_6G_Space-Air-Ground_Integrated_Networks_A_Comprehensive_Tutorial_and_Survey
7. Technical Program - IEEE Xplore https://ieeexplore.ieee.org/iel7/10043349/10043179/10043375.pdf
8. Sensors, Volume 25, Issue 18 (September-2 2025) – 326 articles https://www.mdpi.com/1424-8220/25/18
9. Energies, Volume 16, Issue 3 (February-1 2023) – 546 articles - MDPI https://www.mdpi.com/1996-1073/16/3
10. 333333 23135851162 the 13151942776 of 12997637966 ftp://ftp.cs.princeton.edu/pub/cs226/autocomplete/words-333333.txt
11. AlN: An Engineered Thermal Material for 3D Integrated Circuits https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.202402662
12. Enhanced Thermal Boundary Conductance across GaN/SiC ... https://pubs.acs.org/doi/10.1021/acsami.3c16905
13. [PDF] Origins of heat transport anisotropy in MoTe2 and other bulk van der ... https://www.sciencedirect.com/science/article/am/pii/S2542529323002328
14. Engineering Heat Transport Across Epitaxial Lattice-Mismatched ... https://pmc.ncbi.nlm.nih.gov/articles/PMC10416569/
15. Anisotropic thermal conductivity of AlGaN/GaN superlattices https://iopscience.iop.org/article/10.1088/1361-6528/abc5f2
16. Quantum Noise Suppression at Scale with Crosstalk-Robust Gate Sets https://arxiv.org/html/2603.15758v1
17. Arxiv今日论文| 2026-03-12 - 闲记算法 http://lonepatient.top/2026/03/12/arxiv_papers_2026-03-12.html
18. 学科分类目录 - 中国科学：信息科学 http://scis.scichina.com/ssi.html
19. 机器学习2026_1_13 - arXiv每日学术速递 http://arxivdaily.com/thread/75632
20. Computer Science - arXiv https://arxiv.org/list/cs/new

21. Mathematics, Volume 14, Issue 3 (February-1 2026) – 194 articles https://www.mdpi.com/2227-7390/14/3
22. (PDF) Quantum Learning Under Noise: Toward Trainable Fault ... https://www.researchgate.net/publication/400615051_Quantum_Learning_Under_Noise_Toward_Trainable_Fault-Tolerant_Circuits
23. Federated Learning: Challenges, SoTA, Performance Improvements ... https://ieeexplore.ieee.org/iel8/8782661/10362961/10677499.pdf
24. Superconducting Qubit Decoherence Correlated with Detected ... https://arxiv.org/html/2512.18171v1
25. Two-level systems in superconducting quantum devices due to ... https://www.science.org/doi/10.1126/sciadv.abc5055
26. Synchronous detection of cosmic rays and correlated errors in ... https://www.nature.com/articles/s41467-025-61385-x
27. Mechanically induced correlated errors on superconducting qubits ... https://www.researchgate.net/publication/380490714_Mechanically_induced_correlated_errors_on_superconducting_qubits_with_relaxation_times_exceeding_04_ms
28. Abatement of Ionizing Radiation for Superconducting Quantum ... https://arxiv.org/html/2403.01032v1
29. [PDF] Deliverable – D3.1 Overview of cryogenic integrated and improved ... https://ec.europa.eu/research/participants/documents/downloadPublic?documentIds=080166e5d7ebe4cc&appId=PPGMS
30. CryoCMOS RF multiplexer for superconducting qubit control ... https://www.researchgate.net/publication/402612001_CryoCMOS_RF_multiplexer_for_superconducting_qubit_control_readout_and_flux_biasing_at_millikelvin_temperatures_with_picowatt_power_consumption
31. [PDF] 自动化科学技术名词 - 全国科学技术名词审定委员会 http://www.cnterm.cn/news/tzgg/202503/P020250909521326593979.pdf
32. A comprehensive review of AI-driven approaches for smart grid ... https://www.sciencedirect.com/science/article/pii/S1364032125010974
33. System-Level Optimization of AUV Swarm Control and Perception https://www.mdpi.com/2077-1312/14/4/384
34. (PDF) Quantum Intelligence and the Structure of Reality https://www.researchgate.net/publication/400982795_Quantum_Intelligence_and_the_Structure_of_Reality_Mathematics_Computation_and_Emerging_Physical_Laws
35. Computer Science - arXiv https://www.arxiv.org/list/cs/new?skip=25&show=1000

36. Finite-Time Robust Flight Control of Logistic Unmanned Aerial ... https://www.mdpi.com/2504-446X/8/2/58
37. Arxiv今日论文| 2026-01-14 - 闲记算法 http://lonepatient.top/2026/01/14/arxiv_papers_2026-01-14
38. Issue 1 - Volume 1634 - Journal of Physics: Conference Series https://iopscience.iop.org/issue/1742-6596/1634/1
39. (PDF) AI for Tourism Systems: Demand Forecasting, Experience ... https://www.researchgate.net/publication/402905047_AI_for_Tourism_Systems_Demand_Forecasting_Experience_Personalization_and_Sustainable_Destination_Management
40. Measurement of the top quark pair production cross section in pp ... https://escholarship.org/uc/item/5zk331t3
41. [PDF] In-situ observations of compressible turbulence in planetary ... https://theses.hal.science/tel-01425307v1/file/77457_HADID_2016_diffusion.pdf
42. sqrt{s} - = 7 TeV using the ATLAS Detector at the LHC - ResearchGate https://www.researchgate.net/publication/48410598_Luminosity_Determination_in_pp_Collisions_at_sqrts_7_TeV_using_the_ATLAS_Detector_at_the_LHC
43. Radarconf2016 Program | PDF | Science - Scribd https://www.scribd.com/document/369217611/Radarconf2016-Program
44. Modeling and Field Oriented Control of Salient Pole Wound Field ... https://www.researchgate.net/publication/224134610_Modeling_and_Field_Oriented_Control_of_Salient_Pole_Wound_Field_Synchronous_Machine_in_Stator_Flux_Coordinates
45. Spin Hyperpolarization in Modern Magnetic Resonance https://pubs.acs.org/doi/10.1021/acs.chemrev.2c00534
46. Spin Hyperpolarization in Modern Magnetic Resonance - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC9951229/
47. Arxiv今日论文| 2026-03-25 - 闲记算法 http://lonepatient.top/2026/03/25/arxiv_papers_2026-03-25.html
48. Machine Learning - arXiv https://arxiv.org/list/cs.LG/new
49. 机器学习2025_6_6 - arXiv每日学术速递 http://www.arxivdaily.com/thread/68259
50. Issue 12 - Volume 35 - Measurement Science and Technology https://iopscience.iop.org/issue/0957-0233/35/12
51. A NOISE-ASSISTED DATA ANALYSIS METHOD - DOI https://doi.org/10.1142/S1793536909000047?urlappend=%3Futm_source%3Dresearchgate.net%26utm_medium%3Darticle

52. Sensors, Volume 26, Issue 6 (March-2 2026) – 271 articles https://www.mdpi.com/1424-8220/26/6
53. (PDF) Digital Twin Technology - ResearchGate https://www.researchgate.net/publication/397533044_Digital_Twin_Technology
54. Arxiv今日论文| 2026-03-25 - 闲记算法 http://lonepatient.top/2026/03/25/arxiv_papers_2026-03-25
55. [PDF] 科技论文、专著（译著）统计 - 北邮科研院- 北京邮电大学 https://kyy.bupt.edu.cn/__local/9/2C/97/CF12D4507F2C3EE0AE202F1B919_C40C514E_165506.pdf?e=.pdf
56. 人工智能2025_10_2 - arXiv每日学术速递 https://arxivdaily.com/thread/72283
57. (PDF) The Quantum-Patterned Cosmos [Old] - ResearchGate https://www.researchgate.net/publication/395466257_The_Quantum-Patterned_Cosmos_Old
58. (PDF) Quantum computer error structure probed by ... - ResearchGate https://www.researchgate.net/publication/386592949_Quantum_computer_error_structure_probed_by_quantum_error_correction_syndrome_measurements
59. Arxiv今日论文| 2026-02-06 - 闲记算法 http://lonepatient.top/2026/02/06/arxiv_papers_2026-02-06
60. Mathematics Sep 2018 - arXiv http://arxiv.org/list/math/2018-09?skip=1550&show=2000
61. 2025 Index IEEE Transactions on Consumer Electronics http://ieeexplore.ieee.org/iel8/30/11306167/11339251.pdf
62. [PDF] 北京邮电大学科技论文、专著（译著）统计 https://kyy.bupt.edu.cn/__local/4/90/6B/6474B8057A6E4AB04BD423F82B4_5C411C39_826844.pdf?e=.pdf
63. Entropy, Volume 28, Issue 2 (February 2026) – 120 articles - MDPI https://www.mdpi.com/1099-4300/28/2
64. Reinforcement Learning: An Introduction | Guide books https://dl.acm.org/doi/book/10.5555/3312046
65. Stable approach based diagonal recurrent quantum neural networks ... https://www.nature.com/articles/s41598-026-37973-2
66. A Lyapunov Theory-Based SEIG–STATCOM Voltage Regulation ... https://www.mdpi.com/1996-1073/17/19/4992
67. Hierarchical perturbation compensation system with ERL sliding ... https://www.sciencedirect.com/science/article/abs/pii/S2468601823000184
68. [PDF] arXiv:2304.14435v1 [quant-ph] 27 Apr 2023 https://arxiv.org/pdf/2304.14435
69. Quantum optimal control in quantum technologies. Strategic report ... https://link.springer.com/article/10.1140/epjqt/s40507-022-00138-x

70. On Hyper − Stability Theory Based Multivariable Nonlinear Adaptive ... https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/elp2.70035
71. Trajectory tracking of a quadrotor using a robust adaptive type-2 ... https://www.sciencedirect.com/science/article/abs/pii/S0019057820305668
72. Design of active fault-tolerant control system for Air-fuel ratio control ... https://journals.sagepub.com/doi/10.1177/00368504221094723
73. Quantum secured blockchain framework for enhancing post ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12375084/
74. Enhanced Network Security Protocols for the Quantum Era https://ieeexplore.ieee.org/iel8/49/11151729/11002706.pdf
75. Advancements in secure quantum communication and robust key ... https://www.sciencedirect.com/science/article/pii/S2772918425000062
76. AI-Driven Post-Quantum Cryptography for Cyber-Resilient V2X ... https://arxiv.org/html/2510.08496v1
77. [PDF] Transitioning to Quantum-Safe Cryptography on IBM Z https://www.redbooks.ibm.com/redpieces/pdfs/sg248525.pdf
78. Random-projector quantum diagnostics of Ramsey numbers and a ... https://arxiv.org/html/2508.16699v2
79. Mathematics, Volume 14, Issue 6 (March-2 2026) – 162 articles - MDPI https://www.mdpi.com/2227-7390/14/6
80. Top arXiv papers - SciRate https://scirate.com/?page=17
81. 物理学2025_4_1 - arXiv每日学术速递 https://arxivdaily.com/thread/65797
82. Arxiv今日论文| 2026-01-01 - 闲记算法 http://lonepatient.top/2026/01/01/arxiv_papers_2026-01-01
83. Proceedings of the 36th Chinese Control Conference - IEEE Xplore http://ieeexplore.ieee.org/iel7/8020036/8027314/08029203.pdf
84. Computer Science Aug 2024 - arXiv http://arxiv.org/list/cs/2024-08?skip=7200&show=2000
85. A Survey of Vibe Coding with Large Language Models - arXiv https://arxiv.org/html/2510.12399v1
86. The Multinational Enterprise and the Emergence of the Global Factory https://link.springer.com/content/pdf/10.1057/9781137402387.pdf
87. International Journal of Advanced Research in Management and ... https://www.academia.edu/3063358/International_Journal_of_Advanced_Research_in_Management_and_Social_Sciences_ISSN_2278_6236

88. Dictionary of Acronyms and Technical Abbreviations - Academia.edu https://www.academia.edu/64999604/Dictionary_of_Acronyms_and_Technical_Abbreviations
89. Artificial Intelligence for Quantum Error Correction - arXiv.org https://arxiv.org/html/2412.20380v1
90. [PDF] Artificial Intelligence for Quantum Error Correction - arXiv https://arxiv.org/pdf/2412.20380
91. Reference Architecture of a Quantum-Centric Supercomputer - arXiv https://arxiv.org/html/2603.10970v2
92. [PDF] Reference Architecture of a Quantum-Centric Supercomputer - arXiv https://arxiv.org/pdf/2603.10970
93. Post-Quantum Cryptography and Quantum-Safe Security - arXiv https://arxiv.org/html/2510.10436v1
94. Quantum Meets AI: Pioneering the Future of Intelligent Technology ... https://ieeexplore.ieee.org/iel8/6287639/11323511/11316895.pdf
95. Towards Quantum-Native Communication Systems: State-of-the-Art ... https://arxiv.org/html/2311.05239v4
96. Quantum Technologies for Beyond 5G and 6G Networks - IEEE Xplore https://ieeexplore.ieee.org/iel8/8782661/10829557/11107233.pdf
97. [PDF] AI-Driven Post-Quantum Cryptography for Cyber-Resilient V2X ... https://www.arxiv.org/pdf/2510.08496
98. [PDF] The Nexus of Quantum Technology, Intellectual Property, and ... - arXiv https://arxiv.org/pdf/2602.15051
99. Quantum-Informed Machine Learning for Predicting Spatiotemporal ... https://arxiv.org/html/2507.19861v5
100. Superpositions of Quantum Gaussian Processes - arXiv https://arxiv.org/html/2510.01156v1
101. [PDF] arXiv:2409.01153v1 [quant-ph] 2 Sep 2024 https://arxiv.org/pdf/2409.01153
102. The Halo Architecture for Infinite-Depth Reasoning via Rational ... https://arxiv.org/html/2601.18702v3
103. Fast and Accurate GHZ Encoding Using All-to-all Interactions - arXiv https://arxiv.org/html/2406.10336v1
104. [PDF] Double-bracket quantum algorithms for quantum imaginary-time ... https://arxiv.org/pdf/2412.04554
105. [PDF] High-fidelity Quantum Readout Processing via an Embedded SNAIL ... https://arxiv.org/pdf/2602.14333

106. Quantum spectroscopy of topological dynamics via a ... - arXiv https://arxiv.org/html/2511.23169v1

107. Partial Differential Equations in the Age of Machine Learning - arXiv https://arxiv.org/html/2603.07655v1

108. [PDF] A Linear Algebraic Framework for Dynamic Scheduling Over ... - arXiv https://arxiv.org/pdf/2307.06009

109. Baseband control of single-electron silicon spin qubits in two ... https://www.nature.com/articles/s41467-025-60351-x

110. [PDF] of quasiparticle poisoning in - superconducting quantum circuits - IAEA https://inis.iaea.org/records/4xv6t-4cd82/files/53047082.pdf?download=1

111. [PDF] a new perspective with hole spins in silicon - HAL theses https://theses.hal.science/tel-05284500v1/file/CHAMPAIN_2025_archivage.pdf

112. [PDF] 瀱嗍 /输勃 彩煅 ^敳炊觳Ē Ǎ菝燕嗢惭 #512 - Huawei https://www-file.huawei.com/admin/asset/v1/pro/view/6ca306adab0a4852bacffce25f5573ec.pdf

113. Capstone-Assignment - RPubs https://rpubs.com/BushraT/1075426

114. 333333 23135851162 the 13151942776 of 12997637966 https://www.cs.princeton.edu/courses/archive/spring18/cos226/assignments/autocomplete/testing/words-333333.txt

115. (PDF) The Future Internet - Academia.edu https://www.academia.edu/75876614/The_Future_Internet

116. DynQ: A Dynamic Topology-Agnostic Quantum Virtual Machine via ... https://arxiv.org/html/2601.19635v1

117. A Vehicular-Edge Federated, Quantized YOLOv12 System for Real ... https://ieeexplore.ieee.org/iel8/6287639/11323511/11313047.pdf

118. 2025 Index IEEE Transactions on Biomedical Engineering https://ieeexplore.ieee.org/iel8/10/11245645/11262740.pdf

119. BigBang-Proton Technical Report Next-Word-Prediction Is Scientific ... https://arxiv.org/html/2510.00129v1

120. Digital Twin AI: Opportunities and Challenges from Large Language ... https://arxiv.org/html/2601.01321v1

121. Appl. Sci., Volume 13, Issue 11 (June-1 2023) – 507 articles - MDPI https://www.mdpi.com/2076-3417/13/11?utm_campaign=scinews_appliedsciences_issue_vid&utm_medium=social_corp&utm_source=linkedin&trk=public_post-text

122. Open access articles - Europhysics Letters - IOPscience https://iopscience.iop.org/journal/0295-5075/page/Open-access-articles

123. [PDF] 2025.emnlp-main.0.pdf - ACL Anthology https://aclanthology.org/2025.emnlp-main.0.pdf
124. [PDF] Legal Constructs - ResearchGate https://www.researchgate.net/profile/Marcus-Galdia/publication/346922105_Legal_Constructs_Reflections_upon_Legal-Linguistic_Methodology/links/60741465a6fdcc5f779cfd47/Legal-Constructs-Reflections-upon-Legal-Linguistic-Methodology.pdf
125. [PDF] Intelligent World 2030 - Huawei https://www-file.huawei.com/-/media/corp2020/pdf/giv/2024/intelligent_world_2030_2024_en.pdf?la=en
126. Parity Check Equation - an overview | ScienceDirect Topics https://www.sciencedirect.com/topics/engineering/parity-check-equation
127. HSC Textual Dictionary | PDF - Scribd https://www.scribd.com/document/669514595/Hsc-Textual-Dictionary
128. Qubit error bursts in superconducting quantum processors of ... https://arxiv.org/html/2602.18388v2
129. Distinguishing types of correlated errors in superconducting qubits https://arxiv.org/html/2603.16494v1
130. Electronics, Volume 12, Issue 23 (December-1 2023) – 171 articles https://www.mdpi.com/2079-9292/12/23
131. Energies, Volume 17, Issue 1 (January-1 2024) – 274 articles - MDPI https://www.mdpi.com/1996-1073/17/1
132. Mathematics Feb 2025 - arXiv https://www.arxiv.org/list/math/2025-02?skip=2575&show=2000
133. Machine Learning Feb 2025 - arXiv https://www.arxiv.org/list/cs.LG/2025-02?skip=3050&show=2000
134. Computer Science Jul 2025 - arXiv https://www.arxiv.org/list/cs/2025-07?skip=1275&show=2000
135. Computer Science Oct 2024 - arXiv http://arxiv.org/list/cs/2024-10?skip=2175&show=2000
136. Machine Learning Sep 2024 - arXiv https://www.arxiv.org/list/cs.LG/2024-09?skip=115&show=2000
137. Computer Science Feb 2025 - arXiv https://www.arxiv.org/list/cs/2025-02?skip=10650&show=2000
138. Computer Science Oct 2024 - arXiv http://arxiv.org/list/cs/2024-10?skip=1690&show=2000
139. Computer Science Feb 2022 - arXiv http://arxiv.org/list/cs/2022-02?skip=4635&show=2000

140. Probing Dark Matter-Electron Interactions with Superconducting Qubits https://www.researchgate.net/publication/399521938_Probing_Dark_Matter-Electron_Interactions_with_Superconducting_Qubits
141. Symmetry, Volume 17, Issue 12 (December 2025) – 174 articles https://www.mdpi.com/2073-8994/17/12
142. Electronics, Volume 15, Issue 2 (January-2 2026) – 244 articles - MDPI https://www.mdpi.com/2079-9292/15/2
143. Mathematics, Volume 13, Issue 24 (December-2 2025) – 149 articles https://www.mdpi.com/2227-7390/13/24
144. (PDF) Completeness and Convergence in Discrete Complex Systems https://www.researchgate.net/publication/399827984_Completeness_and_Convergence_in_Discrete_Complex_Systems_A_Comprehensive_Synthesis_of_Lattice_Dynamics_Synchronization_and_Information_Geometry